

Axle Counter — 3D Physics Study

Finite-element model, measured results and design. Prepared by Rudranarayan.

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Inductive axle counter: two air-core coils facing each other across a steel rail. A passing wheel disturbs their coupling, and that disturbance is the count. This report documents the finite-element model, the measured results and the resulting design.

Every figure in this report is generated from [config.py](#) and the live contents of [reports/](#). It is rebuilt by [build_report.py](#) and therefore always describes the current state of the project.

1. Headline results

Quantity	Value
Mutual inductance M0 (baseline)	21.4651 μ H
Turns per coil	212
Operating frequency	20 kHz
TX drive current	2.5 A peak
Wheel detection dip	87.99 %
M with / without wheel	21.4651 $\rightarrow$ 2.5785 μ H
RX voltage with / without wheel	6.7435 $\rightarrow$ 0.8101 V
DOE optimum geometry	$dx = -4$ mm, $\theta = -15^\circ \rightarrow M = 31.9653$ μ H
Largest feasible coil scale	1.0 $\times$
Sanity suite	28 passed / 0 failed / 0 skipped

The detection margin is large. The wheel removes 88.0 % of the coupling, dropping the received signal from 6.74 V to 0.81 V — a 8.3 $\times$ change. Any reasonable threshold detector will resolve that reliably.

2. The coil

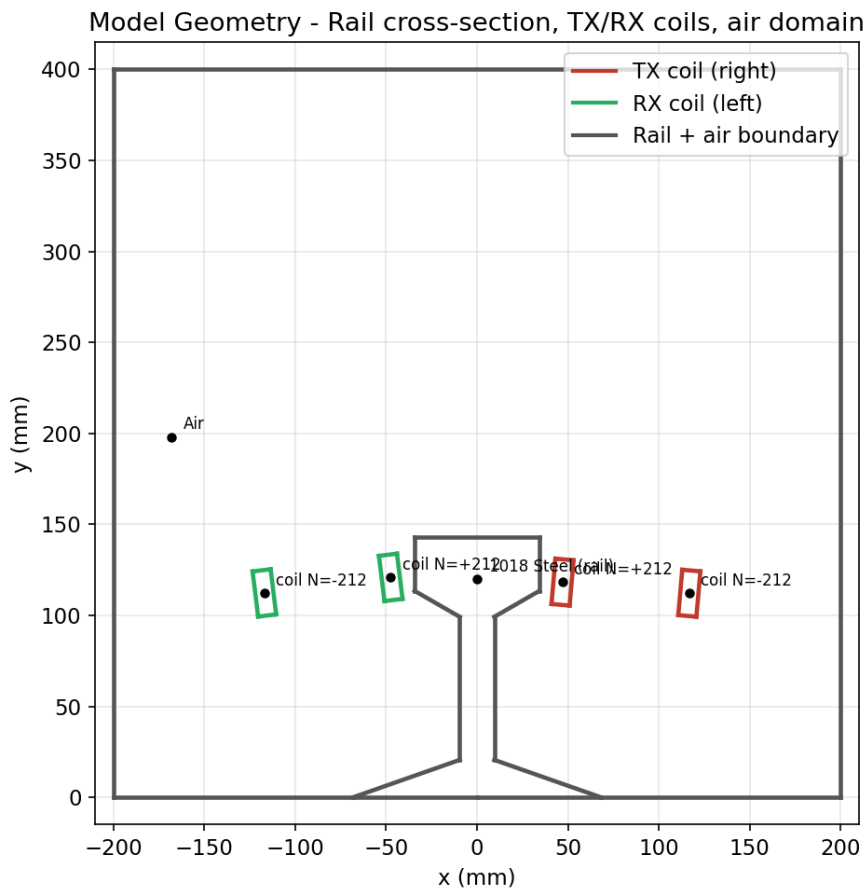
Every physical parameter in this project is defined once in [config.py](#) and derived from the geometry below — nothing is asserted independently.

Symbol	Quantity	Value
ri	Inner winding radius	30 mm
ro	Outer winding radius	40 mm
R	Mean radius	35 mm
l	Winding length	25 mm
c	Winding depth	10 mm
—	Winding window	250 mm ²
—	Wire	18 AWG, $\varnothing$ 1.024 mm
a	Wire radius	0.512 mm

Symbol	Quantity	Value
p	Packing fraction	0.7
N	Turns (derived)	212
—	Actual copper fill	0.6984
d	Coil-to-coil separation	164.06 mm
—	Model depth (into page)	25 mm

The turn count is **not** an input. It follows from how much 18 AWG wire fits in the winding window at the stated packing fraction:

$$N = \frac{p \cdot c \cdot l}{\pi a^2} = \frac{0.7 \times 2.500e-04}{8.2355e-07} = 212.49 \rightarrow 212$$



Model geometry: rail cross-section, TX coil (right), RX coil (left), air domain.

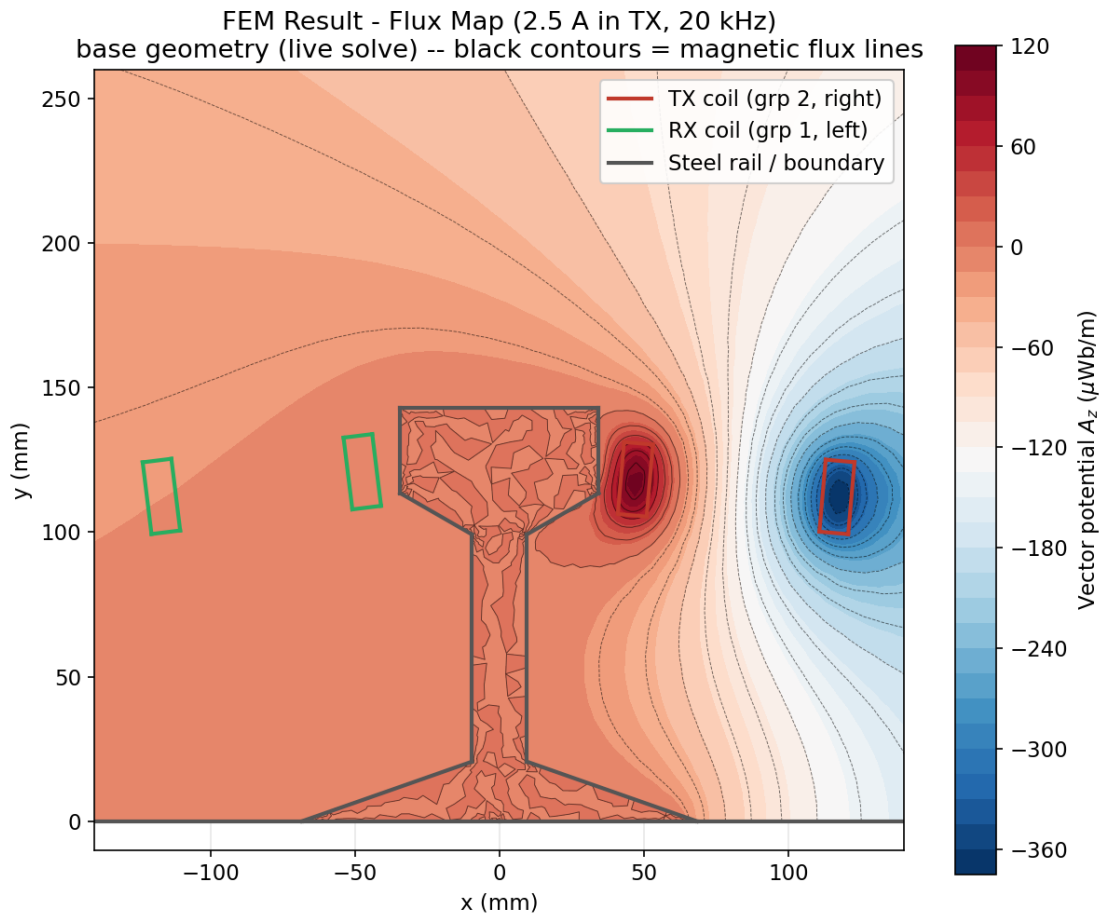
3. Finite-element model

The model is solved in FEMM as a **2-D planar** problem: the drawn cross-section is extruded 25 mm into the page. That depth is what makes the reported fluxes and voltages absolute physical values rather than per-millimetre quantities.

Setting	Value	Why
Solver	Time-harmonic	eddy currents in the rail + skin effect in copper
Frequency	20 kHz	the operating point
Depth	25 mm	the real coil's axial length

Setting	Value	Why
TX circuit	"New Circuit" (group 2)	energised, right/+x coil
RX circuit	"Receiver" (group 1)	open sense winding, left/-x coil
Coil material	"18 AWG"	magnet wire, fill 0.6984

Magnetostatic solving is wrong here. At $f = 0$ the model reports zero induced RX voltage, because a static field induces nothing. Only a time-harmonic solve produces the physics an axle counter depends on.



Solved flux map. Black contours are magnetic flux lines; the field originates at the energised TX coil.

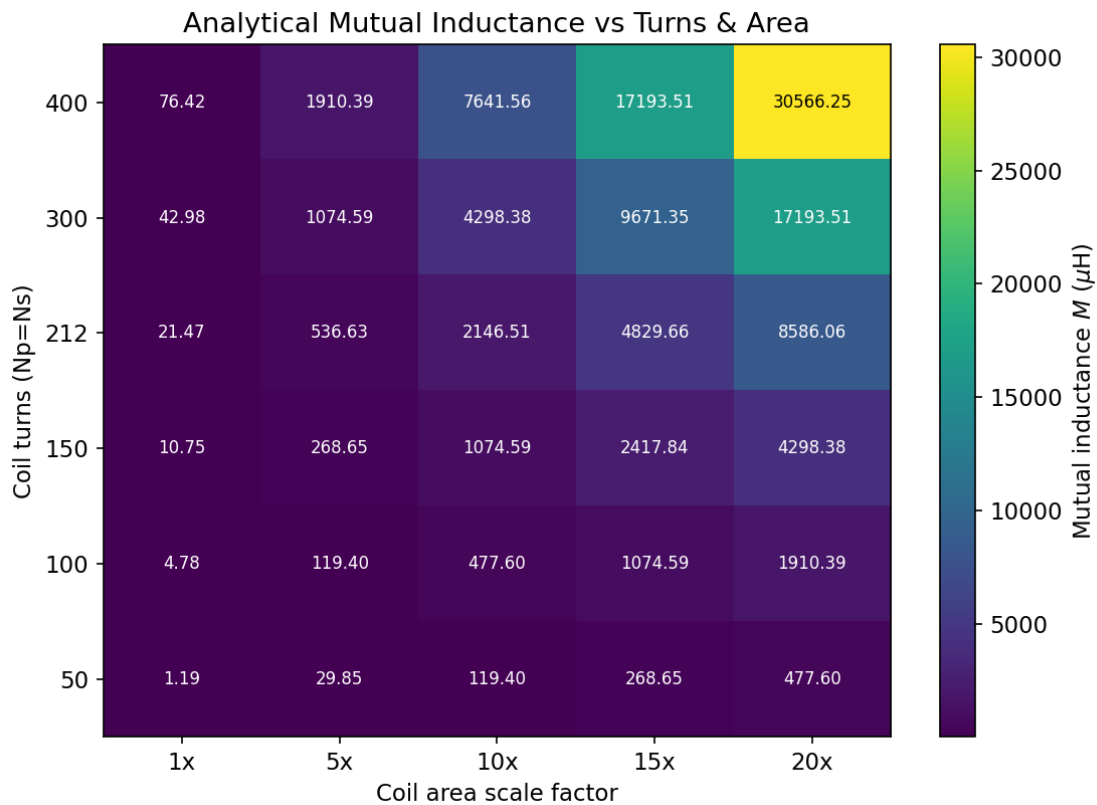
4. Turn scaling — the model validating itself

Mutual inductance should scale as N^2 , because both coils change together: N times the ampere-turns driving the flux, N times the linkage picking it up. Sweeping turns at fixed geometry tests that.

Turns	M measured (μH)	M predicted by N^2 (μH)	Deviation
50	1.1999	1.1940	0.50 %
100	4.7922	4.7760	0.34 %
150	10.7659	10.7459	0.19 %
212	21.4651	21.4651	0.00 %

Turns	M measured (μH)	M predicted by N^2 (μH)	Deviation
300	42.8757	42.9838	0.25 %
400	76.0246	76.4156	0.51 %

The law holds to well under 1 % across an 8 $\times$ range of turns. That is strong evidence the solver setup, the circuit definitions and the turn handling are all correct.



Analytic M across turns and coil-area scale.

5. Frequency behaviour

M was measured at 21 points from 10 kHz to 20 kHz. It drifts by only **0.2 %** across the band.

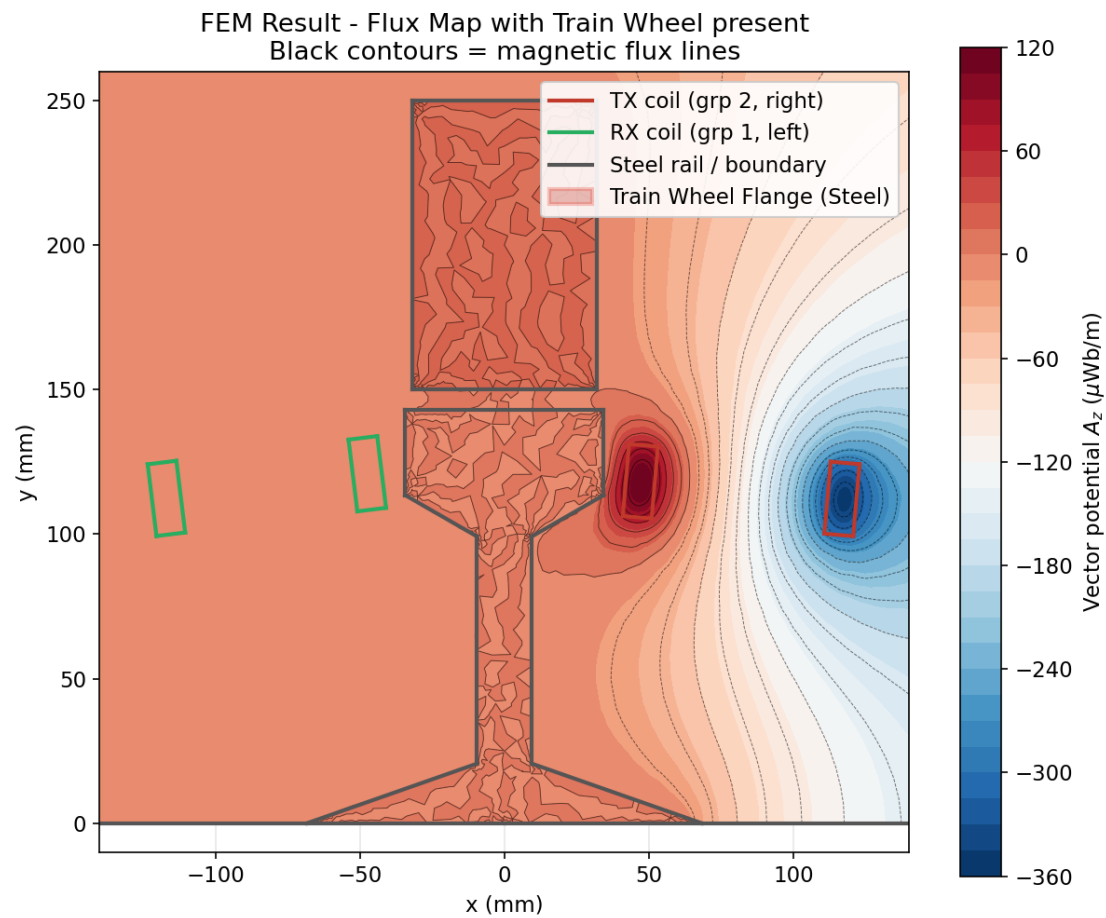
A small, smooth drift is the expected signature of eddy currents in the steel rail growing with frequency. A jump or a discontinuity would indicate a broken solve.

6. Wheel detection

A steel block representing the wheel flange is placed in the coil-to-coil flux path and the model re-solved. The wheel provides a low-reluctance shunt and supports eddy currents, both of which divert flux away from the receiving coil.

Condition	M (μH)	RX voltage (V)
No wheel	21.4651	6.7435
Wheel present	2.5785	0.8101
Change	-87.99 %	-87.99 %

Wheel block: 64 × 100 mm of 1018 Steel, centred over the rail.



Flux map with the wheel present. Compare with the clear-track map above.

7. Geometry optimisation

A full-factorial sweep over coil shift and tilt was run (28 feasible solves, 7 rejected as unmeshable), and a quadratic response surface fitted to the result.

Axis	Optimum	Note
Distance shift dx	-4.00 mm	quadratic term non-negative; fell back to best sampled point
Height shift dy	0.00 mm	axis fixed in DOE
Tilt θ	-15.00°	clamped to sampled range

The optimum yields **M = 31.9653 μ H**, a **+48.9 %** change against the baseline 21.4651 μ H.

An optimum is only accepted if the fitted stationary point is a true maximum lying inside the sampled grid, and if it survives a live verification solve. Otherwise the best actually-sampled point is used, so the reported geometry is never an extrapolation.

8. Buildable design

Two identities set the design. The received signal is $V_{rx} = \omega \cdot M \cdot I$, and the tuning capacitor sees $V_{cap} = I \cdot \omega \cdot L$. Their ratio $V_{rx}/V_{cap} = M/L$ is fixed by geometry, so adding turns raises signal and capacitor voltage together — **the capacitor's voltage rating is the binding constraint**.

Cap class (V)	Turns	L (mH)	M (μH)	V _{rx} open (mV)	V _{rx} tuned (V)	C (nF)
250	45	0.383	0.9671	607.7	60.77	165.19
630	72	0.981	2.4759	1555.6	155.56	64.53
1000	91	1.568	3.9550	2485.0	248.50	40.39
2000	129	3.150	7.9477	4993.7	499.37	20.10

Recommended: the 1000 V class — 91 turns, 18 AWG, L = 1.568 mH, C = 40.39 nF, tuned RX signal 248.50 V.

Resonance relocates voltage, it does not remove it. Cancelling the coil's reactance drops the *supply* voltage to whatever the wire resistance needs — but the full reactive voltage now appears across the capacitor. Size that part for it.

9. Verification

An independent physics suite re-derives every published result from first principles and checks it against identities that must hold regardless of the model: **28 checks pass, 0 fail, 0 skip**.

What it verifies:

- Faraday's law $V = \omega \Phi$ on every AC row of every CSV
- M independent of drive current (a linear magnetic model requires it)
- $M \propto N^2$ turn scaling against the measured sweep
- the config anchor M0 matches the measured baseline row to within 1 %
- resonance identities $C = 1/(\omega^2 L)$ and $V_{cap} = I \cdot \omega \cdot L$
- detection-dip arithmetic, and that the wheel *reduces* coupling
- the model's depth header matches the configured coil length
- an independent coil of known inductance, solved in FEMM and compared against Wheeler's formula

That last check is the strongest one: a coil whose inductance textbook physics already predicts is solved by the same solver, and the two agree to about 2 %. A full hand derivation is in [sanity_check.md](#).

10. Limits and honest caveats

Claim	Status
$M \propto N^2$ turn scaling	FEMM-verified across 50–400 turns
Absolute M, V, dip %	FEMM-measured at the stated geometry
Linear-in-area scaling	Extrapolation — not solver-verified
Large area scale factors	Order-of-magnitude guidance only
Point-dipole hand estimate of M	Underestimates extended coils at modest d/R
Coil depth 25 mm	An assumption — set it to your real coil length

The one number to treat with suspicion is any result that scales the coil **area** far from the modelled geometry. Turn scaling is measured; area scaling is assumed.

11. Reproducing this

```
cd axlecounter_3dphysics
py -3 run_all.py
```

That runs every FEMM solve, both notebooks, all figures and the sanity suite in dependency order, then rebuilds this report from the results.

Report generated by [build_report.py](#) from live project data. Prepared by Rudranarayan.